SQL PORTFOLIO PROJECT

Finance Freight Costs
Freight Distribution & Purchase Price Variance

Oracle Fusion | Accounts Payable | GL Costing | Freight Allocation | PPV

Project Summary

This query identifies freight charges on AP invoices, allocates them proportionally across item lines on the same invoice (weighted by quantity invoiced), then joins cost distributions and standard costs to compute Purchase Price Variance (PPV). It spans AP, GL, Receiving, PO, and Costing modules in a single analytical query.

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1 PROJECT OVERVIEW

Platform:	Oracle Fusion ERP -- AP, GL, RCV, PO, CST modules
Query Type:	Multi-CTE weighted freight allocation + PPV join
Grain:	Cost distribution line / PO receipt line
Key Outputs:	Weighted freight, domestic/foreign cost split, standard cost, variance, GL segments
Parameters:	:PLANT_CODE, :BUSINESS_UNIT_NAME, :P_START_DATE, :P_END_DATE, :LOCATION
Audience:	Finance/Accounting, Supply Chain, Procurement

2 BUSINESS PROBLEM

Freight charges arrive on AP invoices as separate line items not tied to specific purchased goods. Finance needed to allocate those freight costs back to individual item lines (weighted by quantity invoiced), then layer in standard costs to calculate PPV and identify over/under-billed vendors.

- GL account segments had to be filtered to specific freight natural accounts only.
- Domestic vs. foreign currency costs needed to be split for FX exposure reporting.
- Standard cost per unit was required alongside actual cost to compute variance %.
- The query spans AP invoice lines, GL distributions, receipt transactions, and PO headers.
- A CURRENT/PRIOR period flag was derived dynamically from the GL period calendar.

3 TECHNICAL EXPLANATION

- ORG_DATA -- resolves organization and business-unit metadata with optional plant/BU filters.
- STANDARD_COST -- aggregates published standard cost details by item and org, filtered to STATUS_CODE = 'PUBLISHED'.
- CATEGORIES -- pivots up to four procurement catalog levels using conditional MAX() aggregation.
- FREIGHTDISTRIBUTIONS -- isolates freight-type AP invoice lines OR lines hitting specific GL natural accounts.
- ITEMLINES -- collects item-type AP lines to serve as the denominator for proportional weighting.
- TOTALFREIGHTPERINVOICE -- sums total freight per invoice/PO header.
- WEIGHTEDFREIGHT -- allocates freight proportionally: (item_qty / total_invoice_qty) x total_freight. Guards division-by-zero with CASE WHEN SUM != 0.
- Final SELECT -- joins all CTEs; flags rows as CURRENT or PRIOR period using GL_PERIODS.

Key Techniques: Weighted allocation, conditional SUM, GL segment filtering, multi-module joins, currency split CASE, period flag

4 MASKED SQL -- KEY EXCERPTS

```
-- Weighted freight allocation across item lines
WEIGHTEDFREIGHT AS (
  SELECT PO_HEADER_ID, SUM(FREIGHT_AMOUNT) AS FREIGHT_AMOUNT
  FROM (
    SELECT IL.PO_HEADER_ID, IL.INVOICE_ID, IL.QUANTITY_INVOICED,
           TFI.TOTAL_FREIGHT_ON_INVOICE,
           CASE WHEN SUM(AILA.QUANTITY_INVOICED) != 0
                THEN (NVL(IL.QUANTITY_INVOICED,0)
                      / SUM(AILA.QUANTITY_INVOICED))
                    * NVL(TFI.TOTAL_FREIGHT_ON_INVOICE,0)
                ELSE 0
           END AS FREIGHT_AMOUNT
    FROM ITEMLINES IL
    JOIN TOTALFREIGHTPERINVOICE TFI ON TFI.INVOICE_ID = IL.INVOICE_ID
    JOIN AP_INVOICE_LINES_ALL AILA  ON IL.INVOICE_ID  = AILA.INVOICE_ID
    GROUP BY IL.PO_HEADER_ID, IL.INVOICE_ID,
             IL.QUANTITY_INVOICED, TFI.TOTAL_FREIGHT_ON_INVOICE
  ) GROUP BY PO_HEADER_ID
),
-- Domestic vs. Foreign cost split + Standard cost comparison
CASE WHEN UPPER(NVL(TRIM(VC.CURRENCY_CODE),'XXX')) NOT IN ('USD','US')
     THEN NVL(VC.UNIT_COST,0) ELSE 0 END           AS FOREIGN_COST,
CASE WHEN UPPER(NVL(TRIM(VC.CURRENCY_CODE),'XXX')) IN ('USD','US')
     THEN NVL(VC.UNIT_COST,0) ELSE 0 END           AS DOMESTIC_COST,
SC.UNIT_COST                                        AS STANDARD_UNIT_COST,
(SC.UNIT_COST * NVL(RT.PRIMARY_QUANTITY,0))         AS STANDARD_AMOUNT,
CSCV.VARIANCE_AMOUNT,
-- Period flag from GL calendar
CASE WHEN SYSDATE BETWEEN GP.START_DATE AND NVL(GP.END_DATE,SYSDATE+1)
     THEN 'CURRENT' ELSE 'PRIOR' END               AS PERIOD_NAME,
-- GL segment filter (account codes masked)
WHERE GCC.SEGMENT5 IN ([FREIGHT_NATURAL_ACCOUNT_CODES])
  AND (GCC.SEGMENT1 = [COMPANY_CODE_A]
       OR (GCC.SEGMENT1 = [COMPANY_CODE_B]
           AND GCC.SEGMENT3 IN ([LOCATION_CODES])))
  AND GCC.SEGMENT4 = [COST_CENTER_CODE]
```

5 EXAMPLE OUTPUT

Plant	Item #	PO #	Receipt	Std Cost	Actual	Freight	Variance	Period
[PLANT_A]	ITEM-010	PO-55001	R-1293	$2.45	$2.61	$0.08	$0.24	CURRENT
[PLANT_A]	ITEM-011	PO-55002	R-1294	$1.80	$1.75	$0.05	-$0.10	PRIOR
[PLANT_B]	ITEM-012	PO-55003	R-1295	$3.10	$3.22	$0.12	$0.37	CURRENT

** GL account codes, company codes, location codes masked. Values illustrative.*